

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 8

Vectors & Scalar Products

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find magnitude of vector $(3, 4)$ [2 marks]

2. Add vectors $u = (1, 2)$ and $v = (3, 5)$ [2 marks]

3. Find dot product of $a = (2, 3)$ and $b = (1, 4)$ [2 marks]

4. Are vectors $(2, 1)$ and $(4, 2)$ parallel? [2 marks]

5. Find unit vector in direction of $(3, 4)$ [3 marks]

6. If $u = (1, 2, 3)$, find $2u$ [2 marks]

7. Find magnitude of $(3, 4, 0)$ [2 marks]

8. Subtract v from u : $u = (5, 7)$, $v = (2, 3)$ [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Two vectors: $a = (2, 1)$ and $b = (1, 2)$. Find angle between them. (5 marks)

10. If position vector of A is $(1, 3)$ and B is $(4, 7)$, find vector AB. (2 marks)

11. A particle moves with velocity $(2, 3)$ m/s for 5 seconds. Find displacement. (3 marks)

12. Find cross product of $u = (1, 0, 0)$ and $v = (0, 1, 0)$. (4 marks)

13. Three points: A(1, 2), B(3, 5), C(2, 1). Check if they are collinear. (4 marks)

14. A force $F = (3, 4)$ N acts through distance $d = (2, 1)$ m. Find work done ($W = F \cdot d$). (3 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Magnitude of (3,4)

$$||v|| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Q2: Add (1,2) and (3,5)

$$u + v = (1+3, 2+5) = (4, 7)$$

Q3: Dot product (2,3)·(1,4)

$$a \cdot b = 2(1) + 3(4) = 2 + 12 = 14$$

Q4: Are (2,1) and (4,2) parallel?

(4,2) = 2(2,1), YES they are parallel

Q5: Unit vector in direction (3,4)

$$||v|| = 5, \text{ unit vector} = (3/5, 4/5)$$

Q6: Find 2u for u=(1,2,3)

$$2u = (2, 4, 6)$$

Q7: Magnitude of (3,4,0)

$$||v|| = \sqrt{9 + 16 + 0} = \sqrt{25} = 5$$

Q8: u - v for u=(5,7), v=(2,3)

$$u - v = (5-2, 7-3) = (3, 4)$$

SECTION B SOLUTIONS

Q9: Angle between a=(2,1) and b=(1,2)

$$a \cdot b = 4, ||a|| = \sqrt{5}, ||b|| = \sqrt{5}, \cos \theta = 4/(\sqrt{5} \times \sqrt{5}) = 4/5, \theta = \arccos(0.8) \approx 36.87^\circ$$

Q10: Vector AB

$$AB = B - A = (4-1, 7-3) = (3, 4)$$

Q11: Displacement

$$s = v \times t = (2, 3) \times 5 = (10, 15) \text{ m}$$

Q12: Cross product u=(1,0,0) × v=(0,1,0)

$$u \times v = (0, 0, 1) \text{ or k-unit vector}$$

Q13: Check collinearity of A,B,C

$AB = (2, 3)$, $AC = (1, -1)$, $AB \neq k \cdot AC$ for any k , NOT collinear

Q14: Work done $W = F \cdot d$

$$W = (3, 4) \cdot (2, 1) = 3(2) + 4(1) = 6 + 4 = 10 \text{ J}$$